

2520030340

EXPERIMENT – 5

Single and Double Quote Handling in Custom Linux Shell

Aim:

To implement single and double quotes in a custom Linux shell, preserve quoted content and spaces, handle variable expansion, and test edge cases.

Objective:

- To understand single quotes.
- To understand double quotes.
- To preserve spaces inside quotes.
- To handle variable expansion.
- To detect unmatched quotes.
- To test quoted commands.

Theory:

A shell reads commands from the user and executes them.

Single quotes (' ') treat the content as literal text. Variables are not expanded.

Example:

```
echo '$HOME'
```

Output:

```
$HOME
```

Double quotes (" ") preserve spaces and allow variable expansion.

Example:

```
echo "$HOME"
```

Output:

```
/home/kruthi
```

Algorithm:

1. Start the shell.
2. Display the prompt `2520030340_kruthi>`.
3. Read the command.
4. Check for single and double quotes.
5. Preserve the content inside quotes.
6. Do not expand variables inside single quotes.
7. Expand variables inside double quotes.
8. Check for unmatched quotes.
9. Execute the command using `fork()` and `execvp()`.
10. Repeat until `exit` is entered.
11. Stop.

Functions Used:

Function	Purpose
<code>printf(</code> <code>)</code>	Display prompt
<code>fgets()</code>	Read command
<code>fork()</code>	Create process
<code>execvp(</code> <code>)</code>	Execute command
<code>wait()</code>	Wait for process
<code>getenv(</code> <code>)</code>	Get variable value
<code>strdup(</code> <code>)</code>	Store tokens

Sample Input:

```
2520030340_kruthi> echo 'hello world'
2520030340_kruthi> echo '$HOME'
2520030340_kruthi> echo "$HOME"
```

Sample Output:

```
hello world
$HOME
/home/kruthi
```

Observation:

1. Single quotes preserved literal content.
2. Spaces inside quotes were preserved.
3. Variables were not expanded inside single quotes.
4. Variables were expanded inside double quotes.
5. Unmatched quotes were detected.

Result:

The custom shell successfully handled single quotes, double quotes, spaces, variable expansion, and unmatched quotes.

Conclusion:

The experiment successfully implemented **quote handling in a custom Linux shell** and tested different quoted commands and edge cases.

```
kruthi@LAPTOP-LP9QU3RR: ~ × + ▾
command 'car' from deb ucommon-utils (7.0.1-0.1)
command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
command 'phar' from deb php-cli (2:8.2+93ubuntu1)
command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
command 'pchar' from deb pchar (1.5-6)
command 'shar' from deb sharutils (1:4.15.2-9)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `{ '
-bash: !inSingle: event not found
> ^C
kruthi@LAPTOP-LP9QU3RR:~$ nano shell.c
kruthi@LAPTOP-LP9QU3RR:~$ gcc shell.c -o shell
kruthi@LAPTOP-LP9QU3RR:~$ ./shell
2520030340_kruthi> echo 'hello world'
hello world
2520030340_kruthi> echo 'hello      world'
hello      world
2520030340_kruthi> echo '$HOME'
$HOME
2520030340_kruthi> echo "hello world"
hello world
2520030340_kruthi> echo "$HOME"
/home/kruthi
2520030340_kruthi> echo ''

2520030340_kruthi> echo ""

2520030340_kruthi> echo 'hello world'
hello world
2520030340_kruthi> echo "hello world"
hello world
2520030340_kruthi> echo 'hello
Syntax error: unmatched quote
2520030340_kruthi> echo "hello
Syntax error: unmatched quote
2520030340_kruthi> |
```

```

        buffer[j] = '\0';
        args[arg++] = strdup(buffer);
    }
    return 0;ree(args[i]);gs[i] != NULL; i++) {== NULL) { && k <
-bash: syntax error near unexpected token `('
^CCommand 'int' not found, but there are 29 similar ones.
Command 'int' not found, but there are 29 similar ones.
Command 'int' not found, but there are 29 similar ones.
Command 'int' not found, but there are 29 similar ones.
Command 'int' not found, but there are 29 similar ones.
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'pchar' from deb pchar (1.5-6)
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2520030340_kruthi> echo 'hello      world'
hello      world
2520030340_kruthi> echo '$HOME'
$HOME
2520030340_kruthi> echo "hello world"
hello world
2520030340_kruthi> echo "$HOME"
/home/kruthi
2520030340_kruthi>

```